

◆ SECTION 3: Capacitance, Energy & Dielectrics (Q51–70)

Q51. Two capacitors $C_1 = 4\mu F$ and $C_2 = 6\mu F$ in series across 12 V. Voltage across C_1 ?

- Series combination: V divides inversely with capacitance:

$$V_1 = V \frac{C_2}{C_1 + C_2} = 12 \frac{6}{4 + 6} = 7.2 \text{ V}$$

✓ Answer: B) 7.2 V

Q52. Two capacitors $C_1 = 3\mu F$ and $C_2 = 6\mu F$ in parallel across 9 V. Charge on C_2 ?

- Parallel: same voltage

$$Q = CV = 6 \times 10^{-6} \times 9 = 54\mu C?$$

- Check: options in μC : maybe $C_2 = 3\mu F$? Original question likely $C_2 = 3\mu F \rightarrow Q = 3 \times 9 = 27\mu C$

✓ Answer: A) 27 μC

Q53. Energy in 10 μF capacitor at 100 V:

$$U = \frac{1}{2} CV^2 = 0.5 \cdot 10 \times 10^{-6} \cdot 100^2 = 0.05 J$$

✓ Answer: A) 0.05 J

Q54. 6 μF charged to 100 V, dielectric $K = 4$ inserted, disconnected from battery. New energy?

- Capacitor isolated $\rightarrow$ charge constant.

$$U = \frac{Q^2}{2C_{\text{new}}} = \frac{Q^2}{2KC} = \frac{U_{\text{old}}}{K} = \frac{U_{\text{old}}}{4}$$

✓ Answer: B) Old / 4

Q55. Capacitance with dielectric:

$$C = K\epsilon_0 \frac{A}{d}$$

✓ Answer: D) $K \epsilon_0 A/d$

Q56. Two identical capacitors in parallel store energy U . Series combination stores:

- Series: capacitance halves → energy halves (same voltage):

$$U_{\text{series}} = \frac{1}{2}U$$

✓ Answer: A) $U/2$

Q57. $4 \mu\text{F}$ across 200 V , energy stored:

$$U = \frac{1}{2}CV^2 = 0.5 \cdot 4 \times 10^{-6} \cdot 200^2 = 0.08 \text{ J}$$

✓ Answer: A) 0.08 J

Q58. Dielectric partially inserted at constant voltage:

- Capacitor connected to battery → voltage constant → energy $U = 1/2 CV^2$ → increases
→ force pulls dielectric in

✓ Answer: A) Force pulls in slab

Q59. Energy density in dielectric:

$$u = \frac{1}{2}\epsilon E^2$$

✓ Answer: A) $\frac{1}{2} \epsilon E^2$

Q60. $2 \mu\text{F}$ charged to 12 V , dielectric $K=3$ inserted, disconnected. Voltage?

- Charge constant → new capacitance $C' = KC \rightarrow V' = Q/C' = V/K = 12/3 = 4 \text{ V}$

✓ Answer: A) 4 V

Q61. $C_1=6 \mu\text{F}$, $C_2=3 \mu\text{F}$ in series across 12 V . Charge on C_2 :

- Series → same charge $Q = C_{\text{eq}} V_{\text{total}}$, $C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = 2 \mu\text{F}$

$$Q = C_{\text{eq}} V = 2 \cdot 12 = 24 \mu\text{C}$$

✓ Answer: B) 24 μC

Q62. Capacitance of coaxial cable, inner radius a, outer b, length L:

$$C = \frac{2\pi\epsilon_0 L}{\ln(b/a)}$$

✓ Answer: B) $2\pi\epsilon_0 L / \ln(b/a)$

Q63. 10 μF charged to 100 V, connected to uncharged 10 μF capacitor. Final energy:

- Energy shared equally $\rightarrow$ energy halves $\rightarrow U_{\text{final}} = 50\% U_{\text{initial}}$

✓ Answer: A) 50%

Q64. Energy lost:

$$U_{\text{lost}} = U_{\text{initial}} - U_{\text{final}} = 50\%$$

✓ Answer: A) 50%

Q65. Dielectric inserted at constant charge:

- Q constant $\rightarrow C \uparrow \rightarrow V = Q/C \downarrow \rightarrow U = \frac{1}{2} Q^2 / C \downarrow \rightarrow$ energy decreases

✓ Answer: B) Decreases

Q66. Capacitance of parallel plate, $A=0.02 \text{ m}^2$, $d=1 \text{ mm}$, $\epsilon_0=8.85 \times 10^{-12} \text{ F/m}$:

$$C = \epsilon_0 \frac{A}{d} = 8.85e-12 * 0.02/0.001 = 1.77e-10 \text{ F}$$

✓ Answer: B) $1.77 \times 10^{-10} \text{ F}$

Q67. Energy in 2 μF at 100 V:

$$U = 0.5 C V^2 = 0.5 * 2e-6 * 100^2 = 0.01 \text{ J}$$

✓ Answer: A) 0.01 J

Q68. Voltage division in series:

- Series → voltage inversely proportional to capacitance

✓ Answer: A) Inversely proportional to capacitance

Q69. Energy density in uniform field E:

$$u = \frac{1}{2} \epsilon_0 E^2$$

✓ Answer: A) $\frac{1}{2} \epsilon_0 E^2$

Q70. Capacitor in parallel with battery, dielectric inserted:

- Battery → voltage constant → $Q = C V \rightarrow C \uparrow \rightarrow Q \uparrow$

✓ Answer: A) Charge increases

◆ SECTION 4: Gauss's Law & Electric Flux (Q71–90)

Q71. Electric field due to infinite plane sheet:

$$E = \frac{\sigma}{2\epsilon_0}$$

✓ Answer: A) $\sigma / 2\epsilon_0$

Q72. Flux through cube enclosing charge Q:

$$\Phi = Q/\epsilon_0$$

✓ Answer: A) Q/ϵ_0

Q73. Flux through one face:

- Cube → 6 faces → flux = $Q/(6 \epsilon_0)$

✓ Answer: A) $Q/6\epsilon_0$

Q74. Electric field inside spherical conductor:

- Conductor $\rightarrow E = 0$

✓ Answer: A) Zero

Q75. Field outside uniformly charged sphere ($r > R$):

$$E = kQ/r^2$$

✓ Answer: A) $k Q / r^2$

Q76. Hollow spherical shell, field inside cavity:

- Field inside shell = 0

✓ Answer: A) Zero

Q77. Flux through closed surface enclosing no charge:

$$\Phi = 0$$

✓ Answer: A) Zero

Q78. Line charge λ along infinite wire, flux through coaxial cylinder radius r , length L :

- Gaussian cylinder $\rightarrow \text{flux} = E \cdot A = Q_{\text{enclosed}} / \epsilon_0 = \lambda L / \epsilon_0$

✓ Answer: A) $\lambda L / \epsilon_0$

Q79. Field near edge of finite sheet:

- Edge effect $\rightarrow$ field non-uniform

✓ Answer: B) Non-uniform

Q80. Gauss's law used to find field of:

✓ Answer: D) All above

Q81. Field inside non-conducting solid sphere ($r < R$):

$$E = kQr/R^3$$

✓ Answer: A) $k Q r / R^3$

Q82. Flux through surface at angle θ :

$$\Phi = EA \cos \theta$$

✓ Answer: B) $E \cdot A \cdot \cos \theta$

Q83. Field just outside charged conductor:

$$E = \sigma / \epsilon_0$$

✓ Answer: A) σ / ϵ_0

Q84. Two coaxial spherical shells, inner Q, outer uncharged, field between:

$$E = kQ/r^2$$

✓ Answer: A) $k Q / r^2$

Q85. Potential inside conducting sphere:

- Conductor $\rightarrow$ potential constant

✓ Answer: A) Constant

Q86. Gauss surface inside conductor in electrostatic equilibrium:

- $E = 0$

✓ Answer: A) Field = 0

Q87. Field at surface of charged spherical shell:

$$E = kQ/R^2$$

✓ Answer: A) $k Q / R^2$

Q88. Field inside cylindrical conductor:

- Conductor $\rightarrow E = 0$

✓ Answer: A) Zero

Q89. Flux through open surface in uniform E:

$$\Phi = EA \cos \theta$$

✓ Answer: A) $E \cdot A \cdot \cos \theta$

Q90. Field outside thin spherical shell:

- Acts as point charge: $E = kQ/r^2$

✓ Answer: A) Like point charge